KUSHAL PATHAPATI

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AI/ML ENGINEERING | DATA SCIENCE | LLM SYSTEMS

PROFILE

AI and data professional with experience in software quality, financial-data validation and analytics. Builds Python data workflows and LLM applications with careful input validation, traceable evidence and reliable handoffs.

EXPERIENCE

Business & Data Analytics Intern
APR 2026 - JUN 2026

United City Cartage Ltd. | Mississauga, Ontario

- Built a Python PDF extraction pipeline for load-tender information using regex parsing and multi-pass address matching.
- Used RapidFuzz at a 95% threshold to map locations to ERP address IDs; collected Samsara fleet API data and identified duplicate truck records.
- Built Selenium automation that reduced a recurring task from about 30 minutes to under 10 minutes; prepared pivot-table summaries.
- Supported digitizing manual Excel workflows in a web ERP, translated processes into functional requirements, and performed functional testing and defect reporting.

Quality Assurance Analyst
MAY 2022 - AUG 2023

Publicis Sapient | Bangalore, India

- Used SQL and Python to validate financial data supporting investment-portfolio dashboards and reporting.
- Developed Python and Robot Framework automation, reducing manual testing effort by 40%.
- Performed regression testing, documented defects in Jira, and collaborated with developers and business analysts in an Agile environment.

Software Associate
JAN 2020 - MAY 2022

Promaestro Techsource Pvt Ltd | Hyderabad, India

- Automated test cases with Python and Pytest and validated user interfaces with Selenium WebDriver.
- Integrated test workflows into Jenkins CI/CD, reducing deployment time by 30%.

TECHNICAL SKILLS

Programming and analytics: Python, SQL, Pandas, NumPy, R, Excel, Power BI, DAX

Machine learning: scikit-learn, XGBoost, regression, classification, model evaluation

Deep learning and NLP: TensorFlow, neural networks, transformers, NLP, text preprocessing

Generative AI: RAG, embeddings, LangChain, LangGraph, AI agents, MCP, LLM evaluation

Backend and integrations: FastAPI, Flask, REST APIs, Pydantic

Automation and quality: Selenium

SELECTED PROJECTS

FinSight AI - Financial Research Assistant 2026

- Built a Streamlit and FastAPI application using LangGraph to route questions to SEC-filing RAG, MCP market tools or both. The stack includes Pinecone retrieval, Sentence Transformers embeddings, Groq synthesis, and input and output checks.
- Created an SEC HTML loader and a notebook ingestion path for chunking and embedding filing content; retrieval filters indexed evidence by company and filing year. MCP tools cover stock prices, price history and company news.
- I included seven QA cases and evaluation code for faithfulness, relevance and correctness. The compact set checks regressions; filing coverage and external model, vector-store and market-data credentials shape answer quality. Cloud Run may respond more slowly after idle periods.

Aisle Analytics - Retail Analytics & Inventory Optimization JAN 2026

- Built a Python/Pandas workflow for 32M+ Instacart order-product records, including an eight-step cleaning process for missing values, duplicates, range checks and IQR-based outliers.
- Trained an XGBoost reorder classifier across 12.4M user-product pairs. A weighted inventory score combines normalized demand (40%), historical reorder rate (30%) and predicted reorder probability (30%) to create three stocking tiers.
- An n8n workflow export connects to Flask/ngrok endpoints for a Colab pipeline; outputs feed a Power BI dashboard and a Gemini-drafted email report. The on-demand workflow needs an active Colab session and configured credentials.
- Processed 32M+ records in about 20 minutes. The model benchmark is 0.790 AUC-ROC and 69% reorder recall, compared with 15.9% recall for the logistic-regression baseline.

CyberCrime Trends Analysis in Canada NOV 2025

- Integrated and cleaned Statistics Canada cybercrime data from 2015-2024, reshaped it with tidyverse, analyzed time series and year-over-year change, and built Power BI views with DAX and embedded R visuals.
- The analysis found roughly 300% growth in total violations from 2015-2024, led by fraud, with Ontario, British Columbia and Alberta among the leading provincial contributors. This is descriptive analysis, not predictive modeling.
- The analysis follows Statistics Canada definitions and reporting practices; population-adjusted rates would support stronger provincial comparisons. Forecasting was outside the project scope.

EDUCATION

Postgraduate Diploma, Business Insights and Analytics | Humber Polytechnic, Toronto, Ontario | Completed 2026

Bachelor of Technology, Computer Science | Reva University, Bangalore, India | 2016-2020

CERTIFICATIONS

Microsoft Certified: Azure AI Apps and Agents Developer Associate (AI-103)

Microsoft Certified: Azure Data Fundamentals (DP-900) | Microsoft Certified: Azure AI Fundamentals (AI-900)

DataCamp: Data Analyst in Power BI